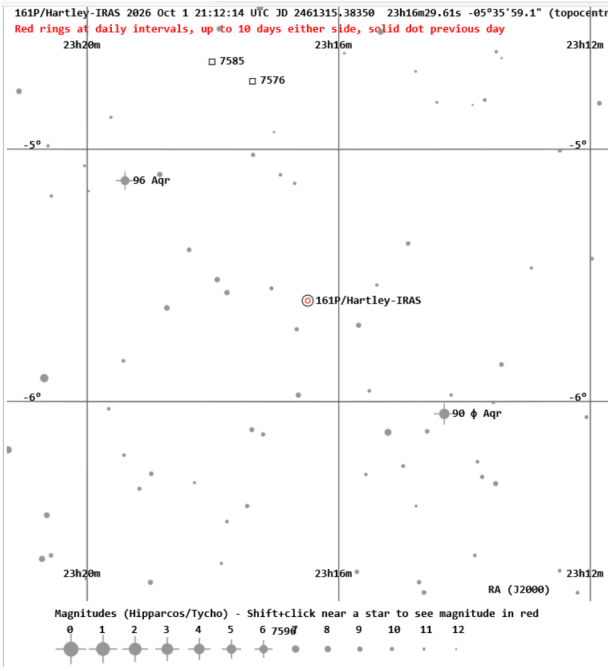
Comets fixed, stars trailing

This document describes a technique for processing a sequence of digital images of a comet. The process will be shown step by step. The software is now all available free online and links will be given. First here is an example of what can be done.

161P Hartley-IRAS captured with Seestar S30 Pro

I noticed on the BAA Comet Section page that this comet might be visible. My own software (grelf.net/ssp) confirmed that it would transit before midnight on October 1 at an altitude of just under 30 degrees from my location. The program produced a finder chart, shown here. I decided to see whether my Seestar could detect the comet.



60 x 30s exposures (in EQ mode) showed no sign of it. So I tried 77 x 60s exposures. My alignment was good: no frames were dropped. In the resulting stack I could just about discern an extremely faint trail where my software said it should be.

I always save the individual FITS frames in the Seestar. After transferring them to my Windows laptop I used some more of my own software to restack keeping the comet fixed while the stars trailed. The result clearly showed the comet among the trails. I then used AstroWizard to smooth the background, reduce noise, and do the standard stretch.



The original stack from the Seestar. Brightness and contrast have been raised as much as possible in Affinity Photo. A very faint trail is just visible.



My restack keeping the comet fixed, so it accumulates in one spot. Enhanced in AstroWizard.

How it was done

What follows is not necessarily for Seestar users only. The basic process and software should work with images obtained from other devices.

Initial considerations

The comet is not visible at all in each of the original images. It only becomes visible when stacked in such a way as to remain stationary.

So the key thing is to work out by how much to shift each frame to trail the stars.

In my example I knew the times of the first and last exposure so in grelf.net/ssp I could find the predicted coordinates of the comet at those times, displayed like this:

```
2026 Oct 1 21:12:14 UTC
JD 2461315.38350
23h16m29.62s -05°35'59.1" (J2000.0)
Azimuth = -22.99°, altitude = 27.06°

2026 Oct 1 22:35:37 UTC
JD 2461315.44140
23h15m50.13s -05°32'33.4" (J2000.0)
Azimuth = 0.70°, altitude = 29.46°
```

Note that Seestar file names show local date/time, not UTC. Yet the metadata in each FITS file have UTC for the start of each exposure (keyword DATE-OBS) and also when the image is saved (DATE-EXP).

So we know how far the comet moves across the sky in terms of RA and Dec.

Image shifting must be done in a coordinate system based on the (x, y) positions of pixels whereas the comet's coordinates are celestial Right Ascension and Declination (RA, Dec). So it was necessary to find the transformation between the two coordinate systems. I discovered that this is called a Helmert transformation. In our case it is complicated by the fact that unlike a normal x-coordinate, RA increases from right to left. Also computers measure the vertical y-coordinate in an image from top to bottom but Dec increases upwards.

To establish the transformation I used reference stars around the comet that could be located in my images.

This led to me developing a new program called AstroFreeze (for freezing comet positions). It is available free as a web page at grelf.net/af. It has a detailed explanation in its own page.

Software used

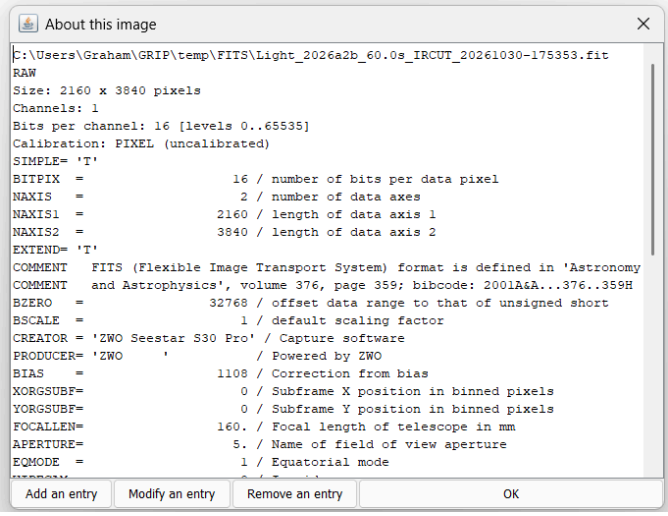
Two programs have been mentioned already:

1. **Solar System Positions** - a free web page at grelf.net/ssp. Set date and time, add a comet to the display, click near it to see its details and generate a finder chart. A graph of distance and possible brightness against date can also be plotted.

2. **AstroFreeze** - a free web page at grelf.net/af. Enter coordinates of 3 reference stars and the predicted comet positions at the first and last times of the sequence of images. Calculate 2 numbers, dx and dy, for input to the stacking process.

3. GRIP (GR's Image Processor) is a Java program. It is also completely free and can be obtained from this web page: grelf.net/download_app.html . This program is needed for all the other stages of processing to be described. Version 26.11.6 or later.

Step 1 - Establish start and end times



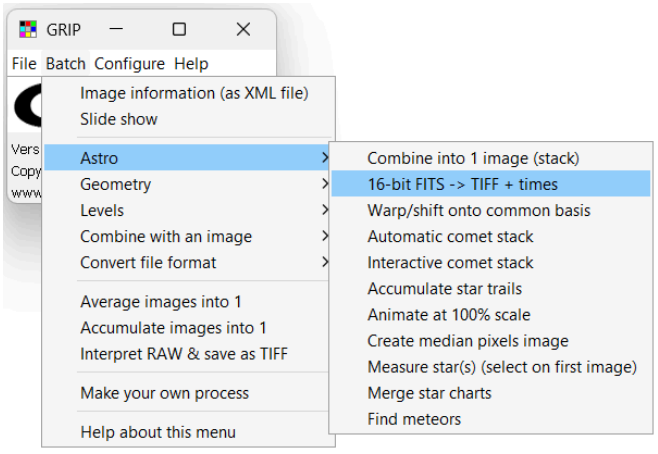
You have a series of images to be stacked in a special way. Look at the metadata in the first and last of the image files to find the UTC times at which they were exposed.

FITS (Flexible Image Transfer System) is a format standard that is maintained by NASA. The first part of a FITS file, its header, is plain text. It can be read by opening the file with a text editor but it will appear messy because although every entry is exactly 80 columns wide there are no line feed characters. GRIP offers a better formatted way to read the headers. From GRIP's **File** menu select **Open image(s)**. You can then use its **Image** menu to **Show image information**.

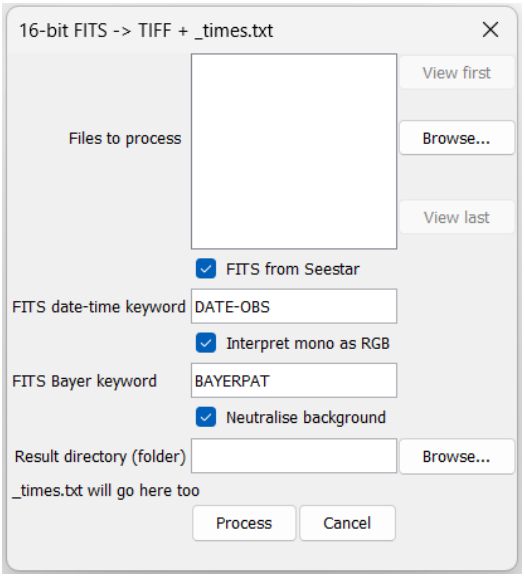
The FITS example shown here reveals how each entry has an 8-character keyword followed by an equals sign and then a value, often followed by a comment. Scroll down to find entries about the time of exposure. FITS files from a Seestar have 2 entries, keywords DATE-OBS and DATE-EXP and they are UTC date/times in a standard format that software can parse.

Step 2 - Convert FITS to TIFF and collect image capture times as numbers

Run GRIP and from the **Batch** menu select **Astro** and then **16-bit Fits → TIFF + times**.



That produces the following dialogue.



Converting to TIFF is useful because the image files can then be opened in a wider variety of programs. But more importantly the process that is started by this dialogue can do additional things.

1. Times can be read from the FITS header. This process saves them in a new file called _times.txt. This file holds simple numbers which are the formatted UTC date/times converted to Julian Days. These times will be needed in the stacking step later because the images were not necessarily taken at constant intervals. There can be several reasons for that. Seestar rejects exposures that it decides are not good enough. Perhaps stars are trailing slightly or

perhaps it has gone cloudy for a while and there are not enough stars detected in the frame. Our stacking to hold the comet steady must take account of this. Notice that you can choose the metadata keyword that identifies the time. Some FITS files will use different names for this.

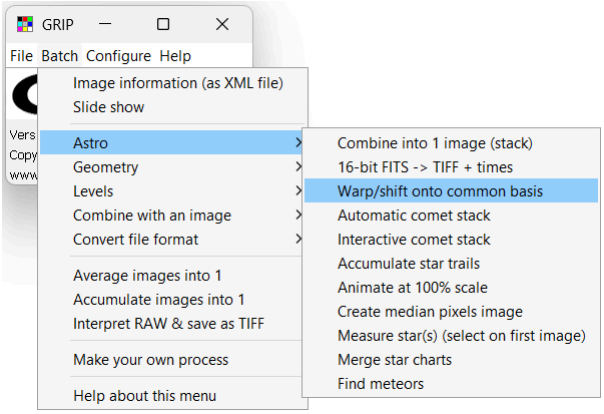
2. Seestar (and other devices) store monochrome images from a detector chip that is covered by a colour filter array (CFA). That needs to be interpreted to make a colour image in red, green and blue (RGB). So this process can optionally do that.

3. While we are doing all this it can be useful to neutralise the background of each image. The program examines the RGB histogram. Background pixels are the most numerous in astro images so there will be a peak towards the left for each colour. The leftmost peak of the three colours is then picked and pixel values in the other 2 colours are stretched so that their peaks coincide with the lowest one. Switch this option off if you wish.

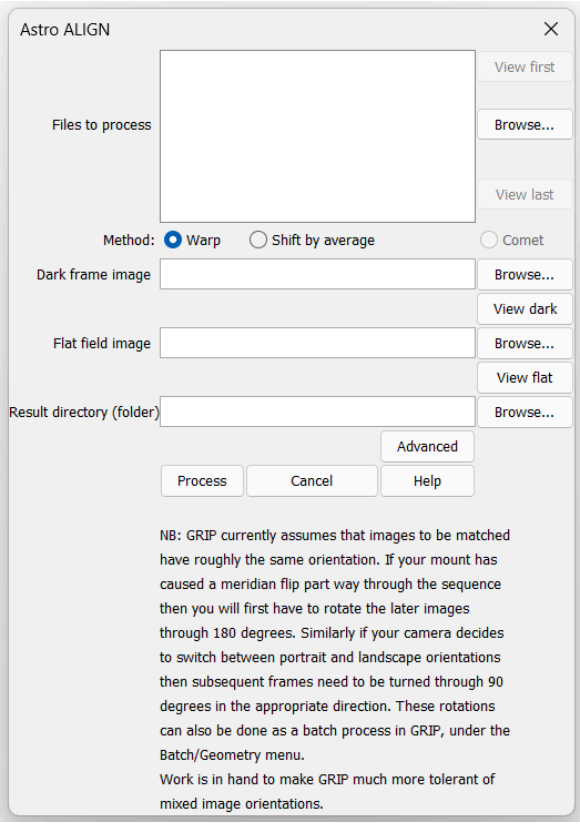
Step 3 - Make new TIFF images that are aligned.

Seestar and other devices dither the images so that stars do not always illuminate the same pixels and any defective pixels have their results smoothed out. In the case of Seestar the device moves slightly after every fifth exposure while it has put a black filter over the detector in order to take a dark frame. Dark frames are used to calibrate the electronic noise levels in the system. Such noise is temperature dependent and so this has to be done during the night time session. It cannot be done before or after that.

So the initial series of images has random positional offsets in it. We must correct that before we can measure star positions.



GRIP has a process for doing this. On the **Batch/Astro** menu again, select **Warp/shift onto common basis** to get another dialogue:

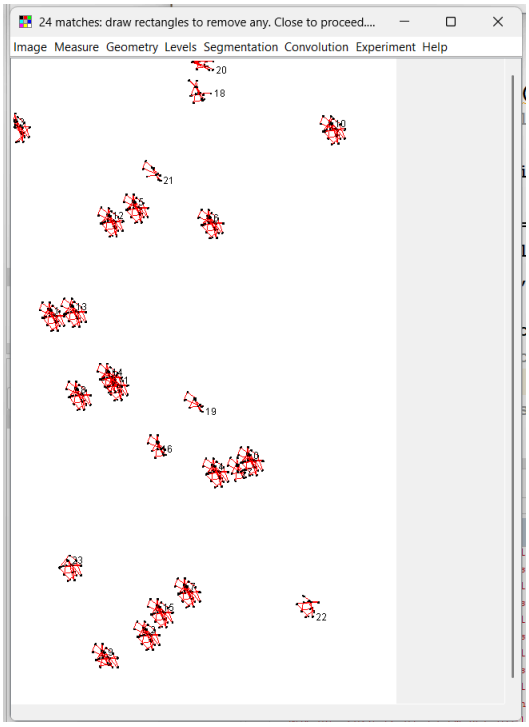


For this purpose you will not need the dark frame or flat frame fields, nor the Advanced button.

The process runs in two passes. On the first pass it measures the brightest stars in each image and analyses how they form a matching pattern between images. The second pass uses that information to make new TIFF files which are aligned with each other: the stars will be in the same positions in every image.

Between passes there is a pause with some displays to allow you to see what is going on and possibly make some amendments.

After pass 1 the first display is like this:



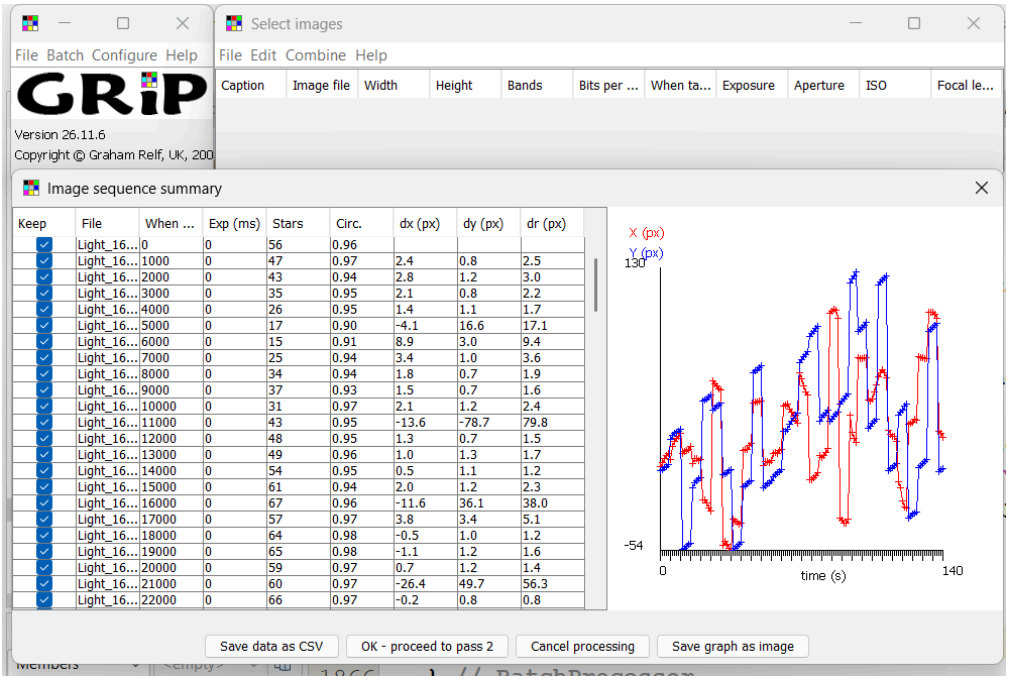
It shows which stars have been used for matching and how they moved around between frames.

If a match looks wrong you can drag a rectangle around it and remove it from the analysis. Possible examples here are stars which have not been used in every image, but these can be kept without any ill effects.

In this particular example I just closed the window in order to proceed, as it says at the top.

Another window then appears, showing details of the measurements that have been made. Of particular interest is the graph on the right which demonstrates how the detector dithered both x and y positions, plotted against frame number.

Proceed to pass 2 by clicking the **OK - proceed** button at the bottom.

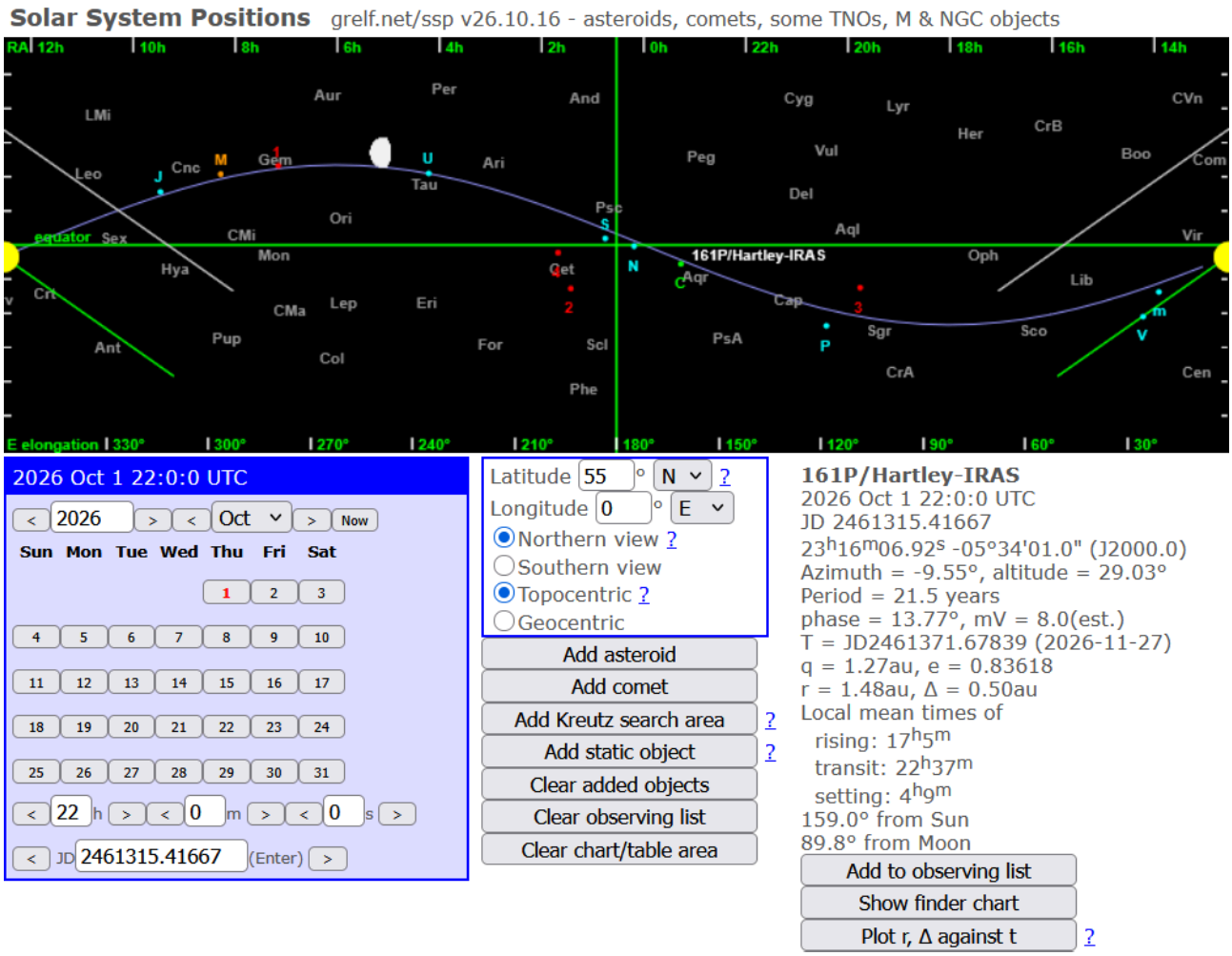


I found it quite interesting to see how my Seestar had jumped around in its dithering.

Step 4 - Measure star positions

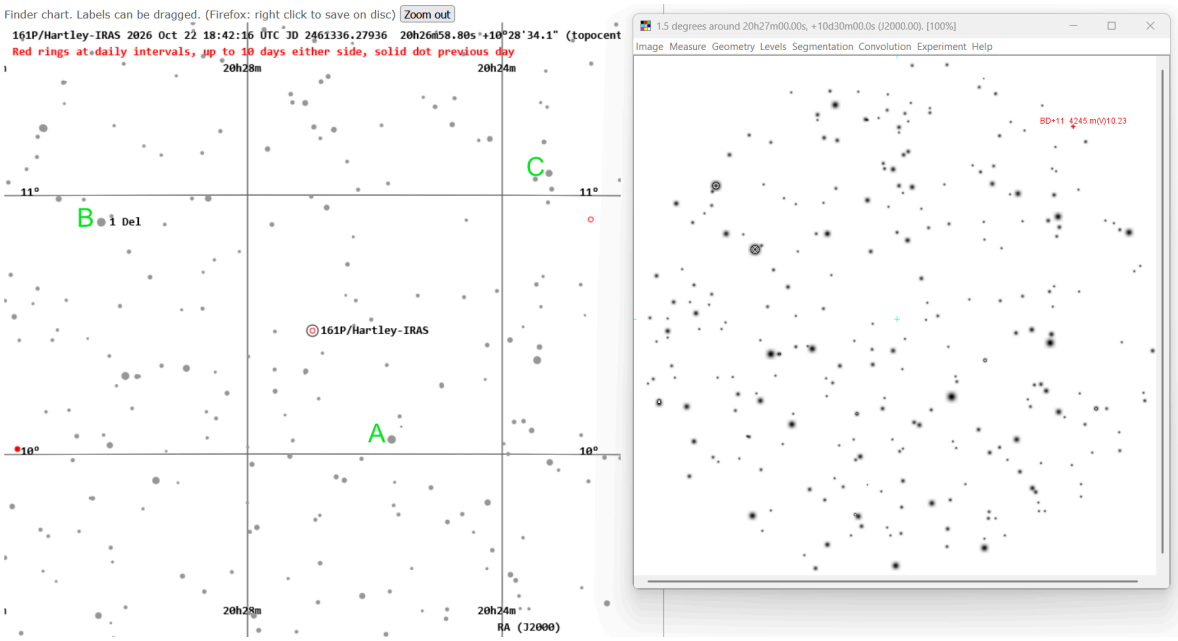
Having aligned the images we need to measure the positions of 3 of the brighter stars. We will need both celestial coordinates (RA, Dec) and positions in any of the images (x, y).

First choose the reference stars. This needs to be done by considering the finder chart around the expected comet position. This chart is created by my Solar System Positions web page at grelf.net/ssp. So I will show more details of that.

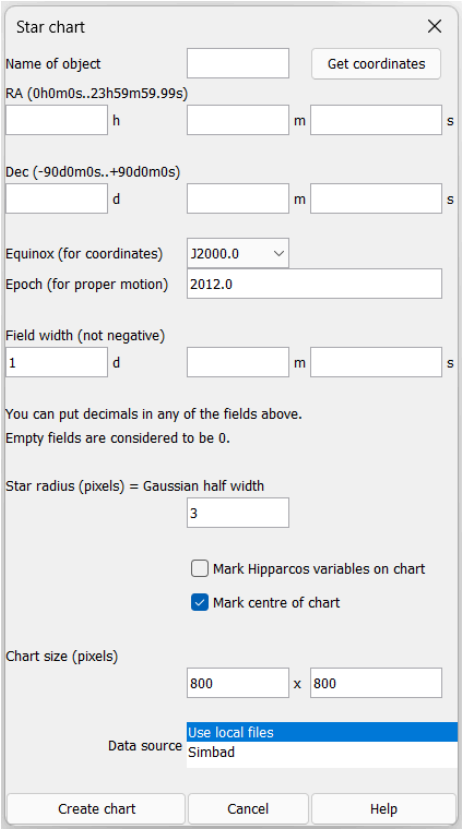


In this example I had set the calendar to a mid-time for the imaging. I used the **Add comet** button to select 161P from a list of periodic comets. Clicking the mouse near it on the upper display gave the details shown on the right. Then I used the **Show finder chart** button to do what that says.

The following example is from a later occasion and I have annotated the chart to show my chosen reference stars. On the right is a second chart from GRIP, to be explained.



We need to know the celestial coordinates of the 3 reference stars and that can be done using GRIP. On the **File** menu select **New star chart** to get a dialogue as shown:



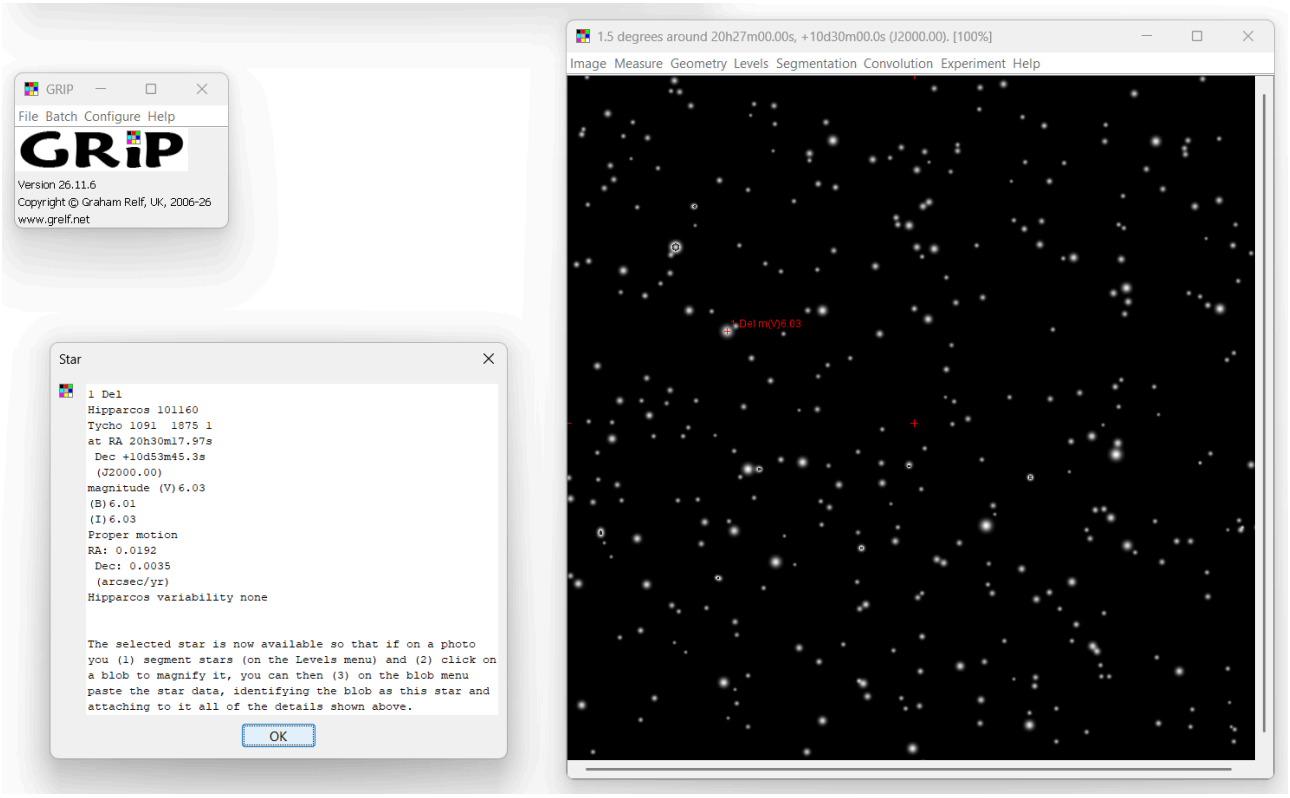
Fill in the RA and Dec fields for the comet position and make the **Field width** 3 degrees. Other fields may be left blank.

There is a choice of using GRIP's own star data (as used to create the finder chart) or asking the online Simbad database (but the October 2026 PHP crisis has made that unavailable, only temporarily I hope).

Click the **Create chart** button.

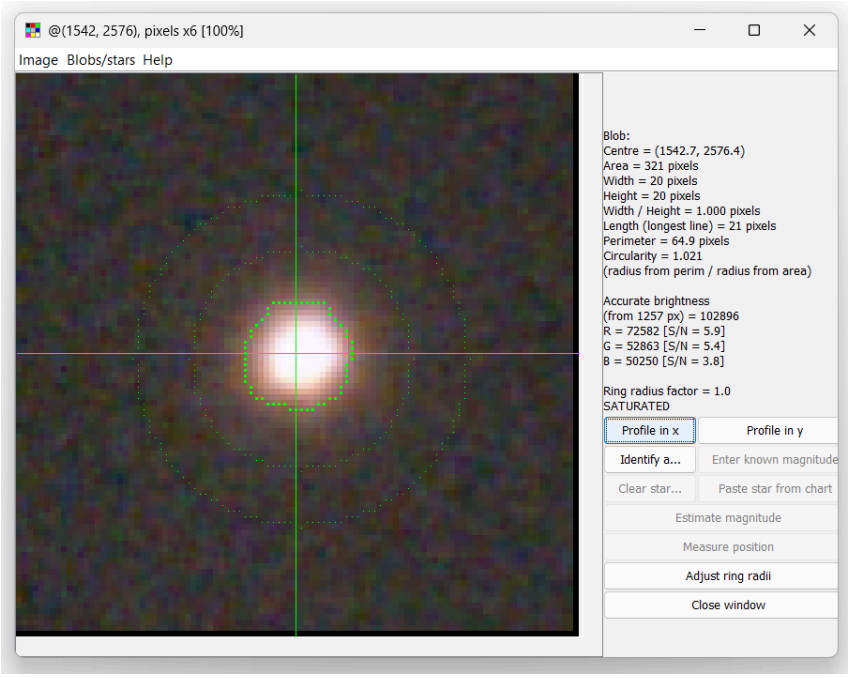
In the example on the previous page I used Ctrl-I to invert the contrast so it looks similar to the finder.

But initially the new chart shows white stars on a black background, as below. In that state use the **Segmentation** menu / **Auto-segment (stars)**. Then it is possible to click on any star to get a window showing its RA, Dec and magnitude:



That can be repeated for the 3 reference stars. Next we need to measure their (x,y) positions in one of the aligned TIFF images.

Open one of those images in GRIP. Auto-segment (stars) as before. This time when you click on a star you will get a different window:



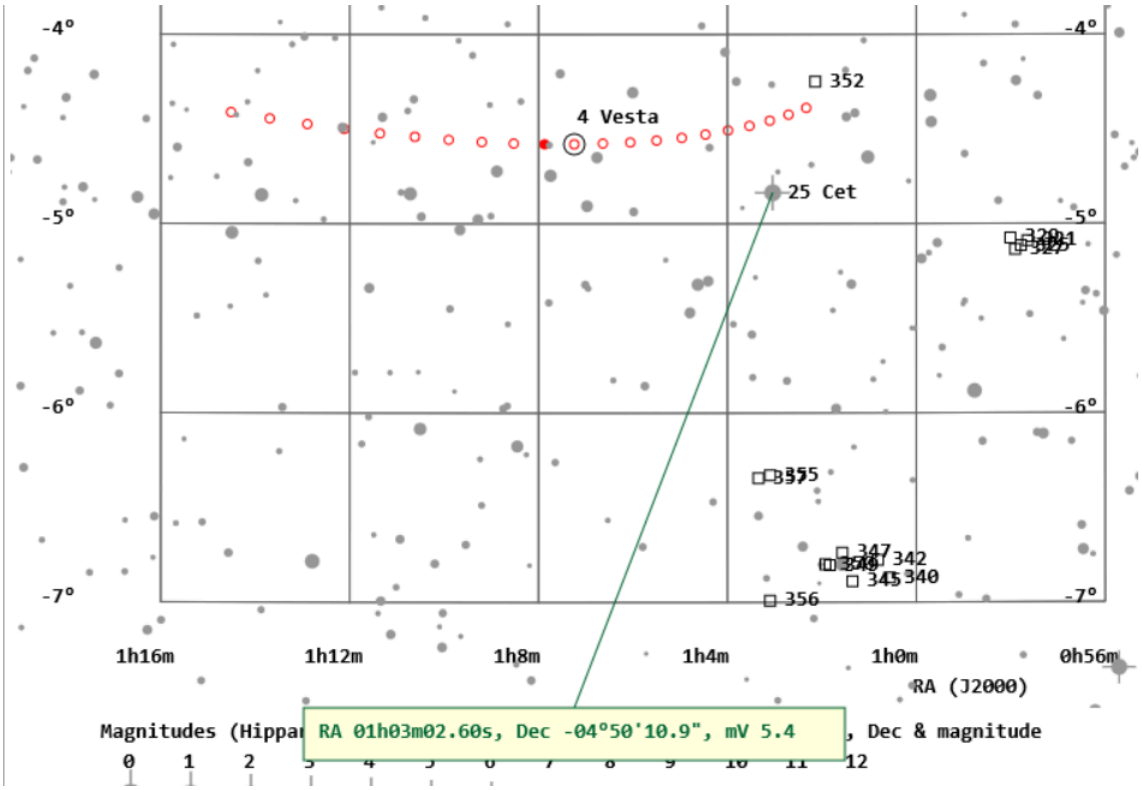
This time the Centre of the detected object is shown, top right, as (x, y).

Get x and y in this way for all 3 reference stars.

To avoid puzzlement: remember that y is measured from top downwards in digital images.

Stop press!

Since writing the section above I have added a new facility to SSP (grelf.net/ssp). Holding the shift key while clicking on a star chart now reveals the RA, Dec and magnitude of the star nearest to the clicked point. Furthermore that string of data is copied to the clipboard as plain text (if your system security permits - it should). This makes it much easier to get the sky coordinates of the reference stars. (Data from Hipparcos. If you want Gaia measurements you will need Simbad, as mentioned above.)



Step 5 - Calculate the Helmert transformation from sky to image

We should now have enough data to complete the table in AstroFreeze (grelf.net/af) and calculate the image offset from start to finish of the sequence.

AstroFreeze v26.10.30

Calculations to enable the position of an asteroid or comet to be held fixed in a sequence of images during stacking, while stars trail past it.

Project name 161P Hartley-IRAS

Ref	Id	RA	Dec	x	y
A	96 Aqr	23h19m23.98s	-05d07m27.7s	300.6	1671.4
B	90 Phi Aqr	23h14m19.36s	-06d02m56.4s	1542.9	2576.6
C	HD 114803	23h15m20.02s	-04d32m03.2s	1295.3	1090.7
start	161P	23h16m29.62s	-05°35'59.1"	1011.31	2136.11
end	161P	23h15m50.13s	-05°32'33.4"	1172.58	2080.24

end - start:
dx = 161.27
dy = -55.88

Load from CSV file | Calculate x & y | Save in browser | Load from browser

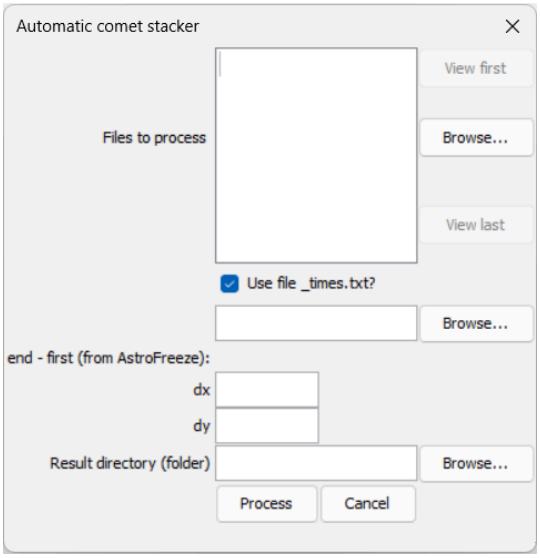
File? Browse... 161P.csv

In this example the **Calculate x & y** button has been pressed, to fill in the lower x and y values and, top right, to show the two numbers we will need for the final stacking.

The AstroFreeze page has further information, including advice about using a CSV file to load the table.

Step 6 - The final stack to keep the comet fixed

Back to GRIP. On the **Batch/Astro** menu select **Automatic comet stack**. The resulting dialogue should be fairly self-explanatory. Using the file *_times.txt* created earlier is optional. If it is not used the images will be shifted by equally spaced amounts and that may not be good enough if the image device did not keep every successive frame.



The result of this step will be saved as *comet.tiff*, in the result directory selected in the dialogue. It will also remain on display in a new window.

My first result was shown on page 1. I had used AstroWizard on the TIFF file from this final step, just to reduce noise and enhance contrast slightly.